

# Ahmed Refat Abd El-Moaty

## Data Engineer | Data Analytics

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### PROFESSIONAL SUMMARY

Results-driven Data Engineer with hands-on experience designing and deploying scalable batch and real-time streaming pipelines, cloud-native data lakehouses, and production-grade analytics architectures. Proficient in Python, PySpark, SQL, Apache Kafka, Apache Airflow, Snowflake, dbt, and AWS/Azure cloud platforms. Skilled in Medallion Architecture (Bronze/Silver/Gold), Apache Iceberg, and dimensional data modeling — Maintaining end-to-end pipeline observability — consistently transforming high-volume raw data into reliable, analytics-ready assets that power business intelligence and machine learning at scale.

### TECHNICAL SKILLS

|                       |                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------|
| Languages & Query     | Python (OOP, Pandas, NumPy, Matplotlib, Seaborn), SQL , Bash / Linux                                      |
| Big Data & Streaming  | Apache Spark, PySpark, Apache Kafka, PySpark Structured Streaming, Hadoop / HDFS                          |
| Cloud Platforms       | AWS (S3, Glue, Redshift, Athena) · Azure (ADLS, Data Factory, Synapse Analytics) · Snowflake · Databricks |
| Data Warehousing      | Snowflake, Amazon Redshift, Azure Synapse Analytics, Apache Iceberg, TimescaleDB                          |
| Data Lakehouse Stack  | Medallion Architecture (Bronze/Silver/Gold), Apache Iceberg, ACID Transactions, Schema Evolution          |
| Orchestration & ETL   | Apache Airflow, dbt, ETL/ELT Pipeline Design, Great Expectations (Data Quality)                           |
| Databases & Formats   | PostgreSQL, MongoDB, SQL Server, Parquet, JSON                                                            |
| DevOps & CI/CD        | Docker, Git / GitHub, Linux                                                                               |
| Monitoring & BI       | Power BI, Grafana, AWS Athena                                                                             |
| Architecture Patterns | Medallion Architecture, Lakehouse, Lambda Architecture, Star / Galaxy Schema, Dimensional Modeling        |

### EXPERIENCE & TRAINING

#### Data Engineer Trainee — Microsoft – Digital Egypt Pioneer Initiative (DEPI) Nov 2025 – May 2026

Cairo, Egypt · Microsoft-Certified Enterprise Data Engineering Program (6 Months)

- Engineered Python and SQL data pipelines across 2+ cloud-based projects on Azure, applying ETL/ELT patterns to ingest, transform, and deliver analytics-ready datasets — reducing data preparation time by 40% compared to manual processing baselines.
- Implemented Star and Galaxy Schema dimensional models across 3+ analytical domains, improving query structure consistency and cutting analytical query complexity by over 35%.
- Completed a 6-month full-lifecycle program across 4 core engineering pillars — ingestion, transformation, warehousing, and orchestration — graduating with a Microsoft Data Engineer Certificate and delivering 2+ end-to-end cloud projects.

#### Cloud Services Trainee — National Telecommunication Institute (NTI) Jan 2026 – Mar 2026

Egypt · Hands-On AWS Cloud Infrastructure Program (2 Months)

- Provisioned and configured 6+ AWS services — EC2, Identity & Access Management , Redshift, Glue, RDS, DynamoDB, and S3 — across 10+ lab environments, building operational readiness for cloud-native data engineering workflows.
- Applied Linux system administration and network fundamentals across 2 months of structured lab practice, reinforcing infrastructure skills critical for managing distributed data systems on AWS.

### TECHNICAL PROJECTS

#### Financial Data Platform: AWS Cloud Data Warehouse [GitHub ↗](#)

AWS Glue (PySpark) · Amazon S3 · Amazon Redshift · Apache Airflow · dbt · Great Expectations · Power BI

- Architected a full Medallion Architecture (Bronze → Silver → Gold) ELT pipeline on AWS, processing 2M+ financial transaction records end-to-end — from raw S3 ingestion through fraud-detection-ready Gold layer datasets.
- Enforced data quality at the Silver layer using Great Expectations, automating schema validation, completeness checks, and accuracy assertions — preventing corrupt data from propagating to downstream analytics.
- Modeled a Star Schema data warehouse in Amazon Redshift using AWS Glue (PySpark) for transformation and dbt for business logic standardization, enabling structured fraud pattern analysis across transaction dimensions.

- Delivered a Power BI fraud detection dashboard connected to Redshift, with end-to-end pipeline orchestration via Apache Airflow DAGs ensuring reliable, scheduled data refresh.

### NYC Smart Traffic: Batch & Streaming Analytics Platform [GitHub ↗](#)

Apache Kafka · PySpark Structured Streaming · Apache Airflow · Snowflake · TimescaleDB · Power BI · Grafana · AWS S3

- Engineered 2 fully independent data pipelines processing 40M+ records: a batch pipeline ingesting NYC traffic, weather, and incident data into a 3-layer S3 data lake (Raw → Processed → Curated) using PySpark, orchestrated by Apache Airflow.
- Modeled a Galaxy Schema data warehouse in Snowflake spanning 3 analytical domains (traffic, weather, incidents), delivering multi-dimensional historical analytics via Power BI dashboards.
- Built a parallel real-time streaming pipeline consuming live Citi Bike station data through Kafka → PySpark Structured Streaming → TimescaleDB, with a live Grafana dashboard tracking station availability and usage in real time.

### Predictive Telecom Streaming Lakehouse [GitHub ↗](#)

Apache Iceberg · AWS S3 · Apache Kafka · PySpark Structured Streaming · Anomaly Detection (Isolation Forest) · Apache Airflow · Grafana · AWS Athena

- Architected a production-grade streaming Data Lakehouse on Apache Iceberg / AWS S3, ingesting 4 concurrent API streams (system, network, radio, environmental metrics) into dedicated Kafka topics for parallel real-time processing.
- Implemented a Bronze → Silver → Gold Medallion pipeline using PySpark Structured Streaming — Silver layer joins all 4 streams into unified Iceberg records on S3 with full ACID support, schema evolution, and automated compaction.
- Deployed an anomaly detection model (Isolation Forest) on each Gold layer micro-batch; decoupled a Python microservice consuming a dedicated Kafka topic to trigger Telegram Bot alerts — fully isolated from the core ML pipeline.
- Orchestrated 2 Airflow DAGs for periodic Iceberg compaction and anomaly model (Isolation Forest) retraining on historical lakehouse data, with live observability via Grafana connected to AWS Athena over the Gold layer.

### Flight Delay Prediction Platform [GitHub ↗](#)

Apache Kafka · PySpark Structured Streaming · TimescaleDB · Grafana · XGBoost · Apache Airflow · Snowflake · Power BI

- Built a unified Streaming + Batch platform integrating 2 live APIs (OpenSky flight data, Open-Meteo weather) — streaming path routes data through Kafka → PySpark Structured Streaming → TimescaleDB → Grafana for real-time telemetry monitoring.
- Embedded an XGBoost Classifier directly into the streaming micro-batch pipeline for real-time flight delay prediction at 82% accuracy — delivering live ML inference alongside raw telemetry without a separate serving layer.
- Constructed a parallel batch pipeline with a 6-step Airflow DAG (Bronze → Silver → Gold → Snowflake), modeling a Galaxy Schema data warehouse and delivering 2 Power BI dashboards for historical delay pattern analysis.

## CERTIFICATIONS

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|-----------------------------------------------------------------------------------------------|----------|
| <b>AWS Certified Solutions Architect</b> · Amazon Web Services                                | Apr 2026 |
| <b>Microsoft Data Engineer Certificate</b> · Digital Egypt Pioneer Initiative (DEPI)          | May 2026 |
| <b>AWS Cloud Foundations</b> · Amazon Web Services                                            | Apr 2026 |
| <b>Cloud Services Management &amp; Operation</b> · National Telecommunication Institute (NTI) | Mar 2026 |
| <b>Linux Command Line &amp; Docker</b> · Udemy                                                | Jan 2026 |
| <b>Analyzing &amp; Visualizing Data with Power BI</b> · Udemy                                 | Dec 2025 |
| <b>Introduction to MongoDB</b> · MongoDB University                                           | Sep 2025 |

## EDUCATION

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|--------------------------------------------------------------------------------------------------|---------------------|
| <b>B.Sc. Computer Science &amp; Information Systems</b>                                          | Graduated: May 2023 |
| Kafr El-Sheikh University, Faculty of Computers and Information · Information Systems Department |                     |

## LANGUAGES

**Arabic** (Native) • **English** (Professional Working Proficiency — B2+)